

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: MLA03 COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Case Study

Duration: 2hrs

A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision

Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior.

(5 Marks)

A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and

reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation.

(5 Marks)

3. An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework

by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. **(5 Marks)**

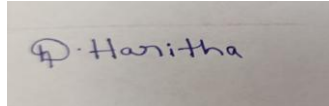
4. A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. **(5 Marks)**
5. A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. **(5 Marks)**
6. A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. **(5 Marks)**
7. A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance. **(5 Marks)**
8. A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions. **(5 Marks)**
9. A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations. **(5 Marks)**
10. A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this

context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. **(5 Marks)**

Code of Conduct:

*I **Dasari Haritha(192425254)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

A photograph of a handwritten signature in blue ink on a light-colored surface. The signature reads "D. Haritha".

Signature of the student:

ASSESSMENT TOOL 2

REINFORCEMENT LEARNING

CASE STUDY

Question 1

Case Study

A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays.

Answer

Understanding of the Problem

The warehouse robot must automatically move products from one place to another while completing tasks quickly, accurately, and safely. Reinforcement Learning (RL) allows the robot to learn the best actions by interacting with the warehouse environment and receiving rewards.

MDP Components

State (S)

The state represents the robot's current situation.

Examples:

- Robot location
- Item location
- Destination location
- Battery level
- Obstacle positions
- Current load
- Traffic inside warehouse

Action (A)

Possible actions include:

- Move Forward
- Move Backward
- Turn Left
- Turn Right

- Pick Item
- Place Item
- Recharge Battery
- Stop

Reward (R)

Reward should encourage efficiency and safety.

Example Reward Function:

Situation	Reward
Pick correct item	+20
Deliver item successfully	+50
Fast delivery	+15
Avoid obstacle	+10

Situation	Reward
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Collision	-50
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Wrong item picked	-40
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Battery exhausted	-30
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Delay in delivery	-15
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Policy (π)

The policy decides which action the robot should take in every state.

Example:

"If an obstacle is detected, change direction before moving."

Environment

The environment includes:

- Warehouse layout

- Shelves
- Goods
- Obstacles
- Charging station
- Workers
- Delivery points

The robot continuously interacts with this environment.

Analysis of Reward Function

The reward function balances three important objectives.

1. Speed

Positive rewards for completing tasks quickly reduce delivery time.

2. Accuracy

Large rewards for selecting the correct item improve delivery accuracy.

3. Safety

Heavy penalties for collisions encourage safe navigation.

Effect of Reward Design

A well-designed reward function enables the robot to:

- Complete deliveries faster.
- Avoid unnecessary movements.
- Reduce accidents.
- Improve warehouse productivity.
- Learn optimal paths over time.

If rewards only focus on speed, the robot may ignore safety and cause collisions. If rewards only focus on safety, deliveries become slow. Therefore, balancing rewards is essential.

Conclusion

By modeling the warehouse robot as a Markov Decision Process with carefully designed states, actions, rewards, and policy, the RL agent learns efficient, accurate, and safe warehouse operations. A balanced reward function improves learning efficiency and overall warehouse performance.

Question 2

Case Study

A streaming platform wants to use Reinforcement Learning to recommend movies dynamically.

Answer

Understanding of the Problem

The platform must recommend movies that users are most likely to watch. RL helps the recommendation system learn user preferences through continuous interaction.

State Representation

The state includes:

- User age
- Preferred genres
- Previously watched movies
- Watch history
- Viewing time

- Device type
- Ratings given
- Recently searched movies

Action Space

Possible actions:

- Recommend Action movie
- Recommend Comedy movie
- Recommend Drama
- Recommend Thriller
- Recommend Latest release
- Recommend Similar movie
- Recommend Trending movie

Each recommendation is considered one action.

Reward Function

User Activity	Reward
Movie watched completely	+20
User likes movie	+15
Positive rating	+10
User watches again	+10
User ignores recommendation	-5
User skips movie	-10
User unsubscribes	-50

Analysis of Reward Signals

Positive rewards:

- Increase user engagement.
- Improve recommendation quality.

- Increase watch time.

Negative rewards:

- Reduce poor recommendations.
- Prevent repetitive suggestions.
- Improve personalization.

Exploration vs Exploitation

Exploration

The system recommends new or less popular movies.

Advantages:

- Learns new user interests.
- Prevents repetitive recommendations.

Disadvantage:

- User may dislike recommendations.

Exploitation

The system recommends movies similar to previously liked movies.

Advantages:

- Higher satisfaction.
- Better immediate engagement.

Disadvantage:

- Less diversity.

Evaluation

A balance between exploration and exploitation provides the best recommendation system by improving both user satisfaction and long-term engagement.

Conclusion

RL continuously learns user preferences through interaction. A balanced reward system and proper exploration strategy improve personalization, customer satisfaction, and platform retention.

Question 3

Case Study

An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage.

Answer

Understanding of the Problem

The agricultural system must provide the correct amount of water and fertilizer while reducing waste and maximizing crop yield.

State

The state includes:

- Soil moisture
- Soil nutrients
- Weather forecast
- Temperature

- Humidity
- Crop growth stage
- Rain prediction

Actions

Possible actions:

- Increase irrigation
- Reduce irrigation
- Apply fertilizer
- Reduce fertilizer
- Delay irrigation
- No action

Reward Function

Situation	Reward
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Healthy crop growth	+40
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High crop yield	+60
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Water saving	+20
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Fertilizer optimization	+15
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Water wastage	-20
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Crop damage	-60
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Over-fertilization	-30
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Delayed Rewards

Crop growth occurs over weeks or months.

Immediate reward:

- Water applied correctly.

Final reward:

- Harvest yield.

Delayed rewards make learning difficult because today's actions affect future crop production.

Suitable Learning Algorithm

Q-Learning is suitable because:

- Learns optimal actions.
- Handles sequential decision-making.
- Improves policy through repeated interaction.
- Works effectively in changing agricultural environments.

Analysis

The RL system gradually learns:

- Optimal irrigation schedule.
- Proper fertilizer quantity.

- Seasonal weather adaptation.
- Water conservation.

Conclusion

An RL-based agricultural system improves crop productivity while reducing water and fertilizer usage. Delayed rewards require continuous learning, making Q-Learning an effective solution.

Question 4

Case Study

A cybersecurity system uses RL to detect and respond to network attacks.

Answer

Understanding of the Problem

The cybersecurity system must detect malicious activities in real time and respond before attackers cause damage.

State

The state includes:

- Network traffic

- Login attempts
- Failed authentications
- Packet rate
- Source IP
- Destination IP
- Malware alerts
- Firewall status

Environment

The environment consists of:

- Users
- Servers
- Firewalls
- Routers
- Attackers

- Network devices

The environment continuously changes as new attacks occur.

Actions

Possible actions:

- Allow traffic
- Block IP address
- Disconnect user
- Alert administrator
- Start deep inspection
- Update firewall rule

Reward Function

Situation	Reward
Correct attack detection	+40

Situation	Reward
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Prevent attack	+60
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Fast response	+20
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False alarm	-15
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Missed attack	-70
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Service interruption	-30
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Challenges

Sparse Rewards

Serious attacks occur infrequently, so positive rewards are limited, making learning slower.

Real-Time Decision Making

The system must make decisions within milliseconds. Delayed responses can lead to data breaches.

Evaluation

A balanced reward function encourages:

- Fast detection.
- Accurate classification.
- Fewer false alarms.
- Better network security.

Conclusion

RL enables adaptive cybersecurity by learning from network interactions. Proper reward design improves attack detection accuracy while ensuring real-time protection.

Question 5

Case Study

A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing.

Answer

Understanding of the Problem

The company wants to assign drivers efficiently and adjust prices according to demand while ensuring customer satisfaction and fairness.

State

The state includes:

- Customer demand
- Driver locations
- Number of available drivers
- Traffic conditions
- Weather
- Time of day
- Ride requests

Actions

Possible actions:

- Assign nearest driver

- Increase fare
- Decrease fare
- Redirect driver
- Wait for another request
- Offer ride discount

Reward Function

Situation	Reward
Successful ride completion	+40
Reduced passenger waiting time	+20
Efficient driver allocation	+25
Customer satisfaction	+20
Ride cancellation	-30

Situation	Reward
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Long waiting time	-20
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Unfair surge pricing	-40
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Environmental Uncertainty

The environment changes continuously due to:

- Traffic congestion
- Weather changes
- Festivals
- Peak-hour demand
- Driver availability

The RL agent must adapt its decisions based on these changing conditions.

Ethical Concerns

Dynamic pricing may create fairness issues:

- Very high prices during emergencies.
- Reduced affordability for passengers.
- Customer dissatisfaction.
- Unequal access to transportation.

To address these concerns, the system should:

- Set maximum price limits.
- Ensure transparent pricing.
- Balance company profit with customer fairness.

Evaluation

A balanced RL model improves:

- Driver utilization.
- Passenger satisfaction.
- Revenue generation.

- Fair pricing.
- Reduced waiting time.

Conclusion

By modeling ride-sharing as an MDP with well-defined states, actions, and rewards, RL can optimize driver allocation and pricing. Considering environmental uncertainty and ethical constraints ensures efficient, fair, and reliable transportation services.

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

